PASSWORD QUEST · TEACHER MATERIALS · SAMPLE LESSON 1

A strong password from random words

Students find out why some passwords are easy to guess, build a made-up password from drawn words in the Password Workshop, and learn why every account needs a different password.

GRADE	LENGTH	SUBJECT	FORMAT
ages 8–15	45 minutes	computing, form time	whole class, pairs

Learning goals

By the end of the lesson, students can:

- explain why short, predictable passwords (`123456` , a name and a year, a word with a number) are weak,
- create a made-up password from randomly drawn words, at least 15 characters long,
- explain why the same password should not be used for several accounts,
- say who to turn to when they have a problem with a password or an account.

Materials

- Projector and teacher computer. Student computers or tablets (one per pair is enough).
- The Password Quest app: the **Passwords** menu (missions) and the **Password Workshop**.
- Worksheet (page 3), one per student. Scissors if you want to cut out the word cards.

> **Key rule for the lesson:** students never write their real passwords, not in the app and not on paper. Every password in the lesson is a made-up example. Do not judge students by the passwords they use at home.

Lesson plan

Time	Activity	What the teacher and students do
5 min	**Warm-up: a house key**	Ask the class: “What would happen if one key opened your home, the school and your bike lock, and you lost it?” Link it to passwords: a password is the key to an account.
8 min	**Vote: which password can I guess?**	Write on the board: `123456` , `Teddy123` , `Football2026` , `ElephantUmbrellaRocket` . Students raise hands to vote for the easiest and hardest to guess. Name the patterns together: numbers in order, a name and numbers, a favorite thing and a year. Why is the last one strong? It has 22 characters and chance chose the words, not a person.
10 min	**Password Workshop (pairs)**	Students open the **Password Workshop**, draw three word cards and look at the strength meter. Then they try three words of their own. On the worksheet (task 2) they write two made-up passwords and the workshop’s rating. Remind them: chance should pick the words, because people choose predictably (`TableChairLamp` is a poor choice).
12 min	**Missions in the app**	Students play missions in the **Passwords** menu. Suggested level: ages 8–10 *Digital Explorers* (easy), ages 10–12 *Account Guardians* (medium), ages 12–15 *Cyber Detectives* (hard). Fast finishers continue with the next mission. Walk around and ask: “Why is this answer better?”

6 min	**One key for every door**	Short discussion: a game announces a password leak. What happens if I use the same password for my school email? Older students can try the simulation in the mission *One key for every door* (Account Guardians). Takeaway: a different password for every account, with help from a password manager approved by the school or parents.
4 min	**Wrap-up and exit ticket**	Repeat the three rules together (box below). Students complete task 4 on the worksheet and hand it in on the way out.

Three rules to remember:

1. Build a password from **drawn words** (at least 15 characters), each starting with a capital letter.
2. **Every account gets a different password.** No names, birth years or numbers in order.
3. **Never tell anyone** your password. If you have a problem, tell a parent or a teacher.

Adapting for age

Younger students (ages 8–10)

- Do task 2 with cut-out cards (page 3) first, then in the app.
- Talk about the “secret key” of Bot the robot, who guards a digital castle.
- Skip the leak simulation, a conversation about one key for every door is enough.

Older students (ages 12–15)

- Add multi-factor authentication (mission *MFA: the second factor*).
- Discuss why forcing a password change every week leads to patterns like Password1, Password2.
- Explain the order: a passkey (mission *Passkeys*), otherwise a random password from a password manager; only a few passwords by heart, made of four random words.

Things to watch out for

- Students might think passwords from the game are fine to use for real. Stress that **examples from the lesson must not be used for real**; real passwords should be made up at home with a parent.
- If a student says they shared a password or someone got into their account, respond calmly and without shaming. Help them change the password and inform a parent or the administrator according to school rules.
- Student progress in the app is stored only in the browser on that device. The app collects no names or passwords.

WORKSHEET · SAMPLE LESSON 1

A strong password from random words

Name and class: _____

Every password on this sheet is made up. Never write your real password here.

1. Sort the passwords

Circle the rating: **E** = easy to guess, **S** = strong.

Password	Rating	Why?
`123456`	E / S	
`Teddy123`	E / S	
`ZebraTulipBridge`	E / S	
`Football2026`	E / S	
`CactusLampPillow`	E / S	

2. Build a made-up password from random words

Draw cards in the Password Workshop or cut out these. Write two made-up passwords and circle what the workshop showed.

Elephant	**Rocket**	**Cactus**	**Bridge**
Anchor	**Guitar**	**Owl**	**Bicycle**

My made-up password	Workshop rating
	Weak · Could be better · Strong · Excellent
	Weak · Could be better · Strong · Excellent

3. Complete the sentences

a) For every account I use a _____ password.

b) I don't tell my password even to _____.

c) When I have a problem with an account, I tell _____.

d) I don't use my _____ or the _____ I was born in my password.

4. Exit ticket

A friend wants to use the password `Jake2014` everywhere. What would you tell them? Write at least two tips.

_____

_____

_____

FOR THE TEACHER · SAMPLE LESSON 1

Worksheet answers

1. Sort the passwords

Password	Rating	Why
`123456`	E	Numbers in order, the most common password of all.
`Teddy123`	E	A common word and numbers in order, a well-known pattern.
`ZebraTulipBridge`	S	Three random words, 16 characters.
`Football2026`	E	A favorite thing and a year, a very common pattern.
`CactusLampPillow`	S	Three random words, 16 characters.

2. Build a password

Accept any password made of three or more drawn words (at least 15 characters) that the workshop rates *Strong* or *Excellent*. If a student uses related words (for example `DogCatMouse`), ask whether someone could guess them.

3. Complete the sentences

- a) **different / unique** password
- b) **a friend** (not even a best friend or a brother or sister)
- c) **a parent or a teacher** (an adult I trust)
- d) **name** or the **year** I was born

4. Exit ticket (examples of good tips)

- Don't use your name and birth year, they are easy to guess.
- Build the password from drawn words, at least 15 characters.
- Use a different password for every account, otherwise one leak puts them all at risk.
- Never tell anyone your password.

Follow-up activities: homework with parents, "Let's check together whether our family uses the same password in several places" (without writing passwords down); next lesson: *Email Detective* (sample lesson 2).
Expert sources: NIST SP 800-63B-4 (2025), NCSC: Three random words, FIDO Alliance (passkeys).